



Department of Artificial Intelligence and Data Science
Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: B.Tech, V & Artificial Intelligence and Data Science & 'B'

Course Code & Title: CS3551 & Distributed Computing

Date and Time: 15.10.2025 & 1.55 pm to 2.40 pm

Name of the Faculty member: Mrs.C.Usharani

Innovative Practice Description

- **Unit / Topic:** Unit IV / Asynchronous Algorithm for Check pointing and Recovery
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO42
- **Activity Chosen:** Mind Map
- **Justification:**

The **Mind Map activity method** is a powerful and innovative teaching and learning strategy that helps in organizing information visually. Through this activity, their understanding of Asynchronous Algorithm for Check pointing and Recovery and its working principle is strengthened. It frequently starts with a single thought, depicted as an image in the middle of a blank page, to which related images, words, or word fragments are added. It's especially effective in educational settings because it encourages creativity, critical thinking, and a deep understanding of concepts.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**

1. Instructor explained the concept of Asynchronous Algorithm for Check pointing and Recovery in classroom.
2. Based on the discussion, the instructor asked the students to create a mind map.
3. Each student created a mind map on Check pointing and Recovery algorithm.
4. They identify key subtopics and add supporting details
5. Students represent the concepts visually including Central theme, Associations, Curved Lines and Keywords.

CO-PO/ PSO mapping:

CO	PO1	PO2	PO3	PO5	PO8	PSO1	PSO 3
CO4	3	2	2	2	2	3	2

(1 — Low 2 — Moderate 3 — High)

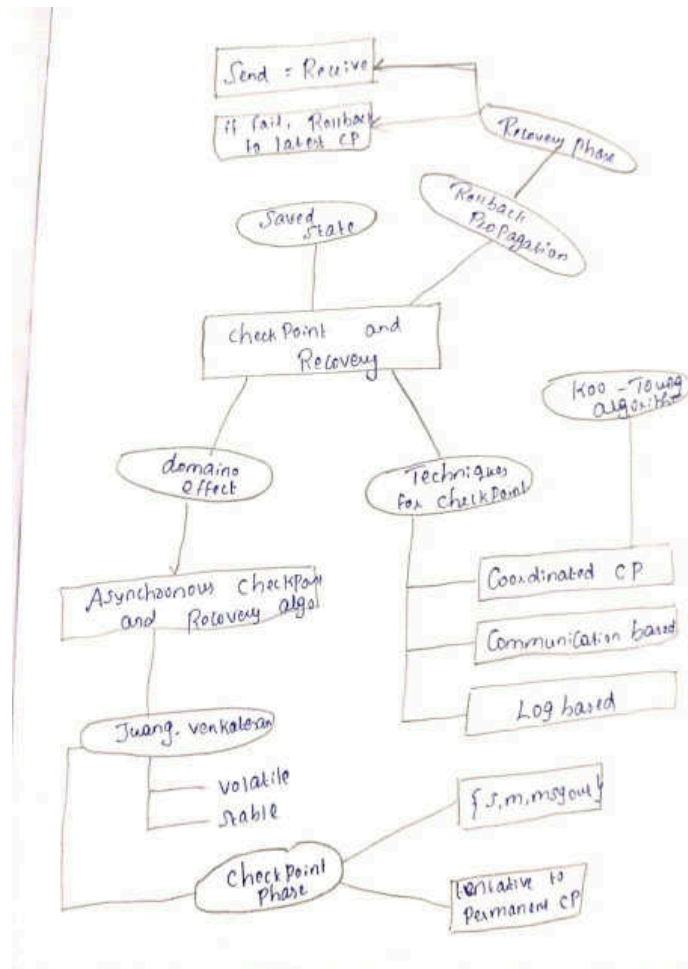
PO & PSO mapping:

Innovative practice	PO1	PO2	PO3	PO5	PO8	PSO1	PSO 3
	3	2	2	2	2	3	2
Justification for correlation	Apply theory and principles of computer science and engineering for Asynchronous Check pointing and Recovery	Identify the algorithms of computer-based system and parameters to for check pointing and Recovery For Asynchrono us system	Able to explore and implement check pointing and Recovery algorithms in distributed systems.	Demonstrate proficiency of adapt tools and techniques for Asynchronou s Check pointing and Recovery	Demonstrate effective communicati on, and solve problems by explaining the Checkpoint and Recovery concept	Students can be able to apply the algorithm for Asynchron ous Check pointing and Recovery to solve a variety of engineering problems particularly in the domain of distributed and parallel computing.	Students can be able to develop the critical thinking skills in checkpoint ing and recovery and provide the solutions to distributed related problems.

- Images / Screenshot of the practice:**



Sample Mind map:



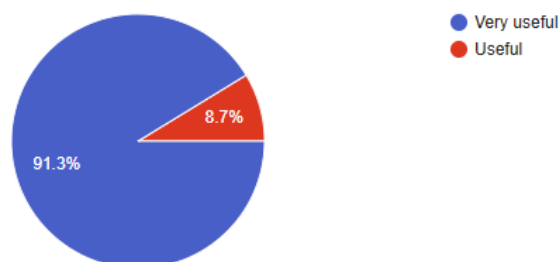
- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students felt the mind map activity engaging and helpful in visualizing key concepts
- ✓ Students appreciated the interactive nature, stating it improved their retention of information

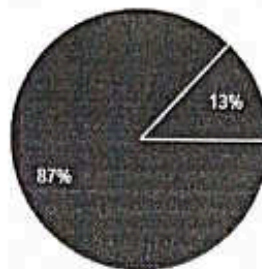
1. How useful was the mind map paper activity that we did in class today?

46 responses



2. Do you have a clear understanding of Asynchronous Checkpoint and Recovery Algorithm?

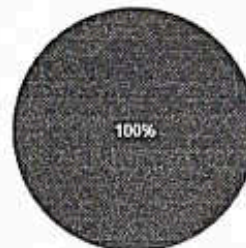
46 responses



● Excellent
● Good
● Fair

3. Does this mind map activity improve listening and grasping capability?

46 responses



● Good
● Fair

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ Students can think outside the box or from diverse perspectives by participating in this activity
- ✓ Improves understanding and helps in breaking down the complex topics into simpler parts

❖ **Challenges faced in implementation:**

- ✓ Some students struggled with organizing their ideas effectively

References:

1. <https://www.mindmapping.com/mind-map>
2. <https://www.mindmapping.com/>

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